AQ2.6: Activity Questions 6 - Not Graded

This assignment will not be graded and is only for practice.
Level 1:

1 point
Let $Ax = b$ be a matrix representation of a system of linear equations. Let $[R|c]$ be the reduced row echelon of the augmented matrix $[A|b]$ corresponding to the system. Choose the set of correct options.

[Hint: Recall reduced row echelon form.]

- [ ] If the system $Rx = c$ has infinitely many solutions, then the system $Ax = b$ has infinite solutions.
- [ ] If the system $Rx = c$ has no solutions, then the system $Ax = b$ has a unique solution.
- [ ] If the system $Rx = c$ has a unique solution, then the system $Ax = b$ has no solution.
- [ ] If the system $Rx = c$ has a unique solution, then the system $Ax = b$ has a unique solution.

No, the answer is incorrect.
Score: 0
Accepted Answers:
If the system $Rx = c$ has infinitely many solutions, then the system $Ax = b$ has infinite solutions.
If the system $Rx = c$ has a unique solution, then the system $Ax = b$ has a unique solution.

Consider a system of linear equations

$$\begin{aligned} 2x_1 + x_2 &= 1 \\ -x_1 + x_3 + x_4 &= -1 \\ x_1 + x_2 - x_3 + x_4 &= 2 \\ -x_1 + x_3 + x_4 &= 1. \end{aligned}$$

If the following matrix represents the augmented matrix of the system, then answer the questions 2,3 and 4.

Find the value of a_{22}.

No, the answer is incorrect.
Score: 0
Accepted Answers:
(Type: Numeric) 0
1 point
Find the sum of the elements of the 4-th row of the augmented matrix.

No, the answer is incorrect.
Score: 0
Accepted Answers:

(Type: Numeric) 2

1 point

Find the sum of the elements of the 3-rd column of the augmented matrix.

No, the answer is incorrect.
Score: 0
Accepted Answers:

(Type: Numeric) 1

1 point

1 point

Let $Ax = b$ be a matrix representation of a system of linear equations and $b = 0$. Choose the set of correct options.

[Hint: Recall the definition of the trivial solution of a system of linear equations and applicability of Gauss Elimination Method.]

- [ ] If A is an invertible matrix then the system has no solution.
- [ ] If A is an invertible matrix then the system has a unique solution.
- [ ] If A is an invertible matrix then the trivial solution is the only solution for the system.
- [ ] If $det(A) = 0$, then the system has infinitely many solutions.

No, the answer is incorrect.
Score: 0
Accepted Answers:

If A is an invertible matrix then the system has a unique solution.
If A is an invertible matrix then the trivial solution is the only solution for the system.
If $det(A) = 0$, then the system has infinitely many solutions.

1 point

Let $Ax = b$ be a matrix representation of a system of linear equations and $b \neq 0$. Choose the set of correct options.

- [ ] If A is an invertible matrix, then the system has no solution.
- [ ] If A is an invertible matrix, then the system has a unique solution.
- [ ] If A is an invertible matrix, then the trivial solution is a solution for the system.
- [ ] If $det(A) = 0$, then either the system has no solution or the system has infinitely many solutions.

No, the answer is incorrect.
Score: 0
Accepted Answers:

If A is an invertible matrix, then the system has a unique solution.
If $det(A) = 0$, then either the system has no solution or the system has infinitely many solutions.

1 point

Consider the following systems of linear equations:

$$\begin{array}{ll} \text{System I:} & \begin{aligned} x - y &= 3 \\ -y + 2z &= 1 \\ x + y + z &= 0 \end{aligned} \\ \text{System II:} & \begin{aligned} x - y &= 3 \\ 2y + z &= 1 \\ 6y + 3z &= 0 \end{aligned} \\ \text{System III:} & \begin{aligned} x - y &= 3 \\ 2y + z &= 1 \\ 6y + 3z &= 3 \end{aligned} \end{array}$$

System I:System II:System III:

x − y = 3−y + 2z = 1x + y + z = 0x − y = 32y + z = 16y + 3z = 0x − y = 32y + z = 16y + 3z = 3

Choose the correct option.

- [ ] All the three systems have a unique solution.
- [ ] System I has a unique solution.
- [ ] System II has no solution.
- [ ] System III has infinitely many solutions.
- [ ] Both the System II and III have infinitely many solutions.
- [ ] Both the system II and III have no solution.

No, the answer is incorrect.
Score: 0
Accepted Answers:

System I has a unique solution.
System II has no solution.
System III has infinitely many solutions.

Level 2:

Suppose P P is a 3×3 3 × 3 real matrix as follows: $\begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix}$ 100010111

The vector X_n Xn is defined by the recurrence relation $PX_{n-1} = X_n$ P Xn−1 = Xn. If $X_2 = \begin{bmatrix} 7 \\ 8 \\ 3 \end{bmatrix}$ X2 = 783, what is the sum of all the elements of X_0 X0?

No, the answer is incorrect.
Score: 0
Accepted Answers:

(Type: Numeric) 6

1 point

1 point

Consider the system of linear equations given below:

$$\begin{aligned} x - y + z &= 2 \\ x + y - z &= 3 \\ -x + y + z &= 4. \end{aligned}$$

x − y + zx + y − z−x + y + z = 2 = 3 = 4.

The system of linear equations has

[Hint: Use the relation between a system of linear equations and determinant of corresponding coefficient matrix.]

- ☐ no solution.
- ☐ infinitely many solutions.
- ☐ a unique solution.
- ☐ finitely many solutions.
- ☐ 3 dependent variable.
- ☐ 2 dependent and 1 independent variable.

No, the answer is incorrect.
Score: 0
Accepted Answers:

a unique solution.
3 dependent variable.

1 point

Consider the system of linear equations:

$$\begin{aligned} -x_1 + x_2 + 2x_3 &= 1 \\ 2x_1 + x_2 - 2x_3 &= -1 \\ 3x_2 + cx_3 &= d \end{aligned}$$

−x1 + x2 + 2x3 = 12x1 + x2 − 2x3 = −13x2 + cx3 = d

Choose the set of correct options.

[Hint: Recall the Gauss elimination method.]

☐

If $c = 2$ and $d = 1$c = 2 and d = 1, then the system has infinitely many solutions.

☐

If $c = 1$ and $d = 1$c = 1 and d = 1, then the system has infinitely many solutions.

☐

If $c = 1$ and $d = 2$c = 1 and d = 2, then the system has no solution.

☐

If $c = 3$ and $d = 2$c = 3 and d = 2, then the system has a unique solution.

No, the answer is incorrect.
Score: 0
Accepted Answers: